RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Third Year - Semester II Examination – February - April 2025

COM 3306 – OPERATING SYSTEMS

Time: Three (03) hours

- Answer **ALL** questions.
- This paper contains **five (05)** questions on **three (03)** pages.

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1. a) List three key components of a computer system. (03 marks)

   b) Differentiate between batch processing and time-sharing Operating Systems (OS). (03 marks)

   c) Explain any two main roles of an OS. (04 marks)

   d) Evaluate the importance of hybrid kernel architecture in modern OS design. (05 marks)

   e) If you are designing a secure OS, what features would you include to enhance security? Justify your choices. (05 marks)

2. a) Describe a Process Control Block (PCB) and list any two types of information it contains. (03 marks)

   b) Distinguish between "CPU-bound" and "I/O-bound" processes. (02 marks)

   c) Explain the difference between long-term, short-term, and medium-term schedulers. (03 marks)

d) The following table indicates the CPU burst time and arrival time for five processes P0, P1, P2, P3, P4, and P5.

Process	Arrival time(ms)	Burst time(ms)
P0	0	5
P1	1	4
P2	1	9
P3	3	5
P4	4	2
P5	5	6

i. Draw the Gantt charts for the following algorithms. (06 marks)

i. First Come First Serve (FCFS)

ii. Shortest Job First (SJF)-Non preemptive

iii. Round Robin (RR) (Time quantum=3ms)

ii. Find the average turnaround time, waiting time, and completion time for each of the above algorithms. (06 marks)

3. a) Define Inter-process Communication (IPC) in the context of computer systems. (02 marks)

b) Explain the difference between direct and indirect message passing. (04 marks)

c) Explain what a critical section is and how the critical section problem is solved in OS. (04 marks)

d) In the Producer–Consumer problem, how would you implement synchronization using semaphores? (04 marks)

e) Assess how the practical limitations of Peterson's solution impact its applicability in real-time systems, where timing constraints are crucial. (06 marks)

4. a) List and explain the four (04) necessary conditions for a deadlock to occur. (04 marks)

b) Compare and contrast the Resource Allocation Graph (RAG) method and the Banker's Algorithm for deadlock handling. (04 marks)

c) Given the following resource allocation state, use the Banker's Algorithm to determine whether the system is in a deadlock state. (04 marks)

Process	Current Allocation (A, B)	Maximum need (A, B)	Current Available (A, B)
P1	(1, 2)	(3, 3)	(1, 1)
P2	(2, 1)	(4, 2)	
P3	(1, 1)	(2, 2)	

d) Analyze the impact of increasing the number of resources on deadlock probability. (04 marks)

e) Analyze how demand paging improves memory efficiency compared to contiguous memory allocation. (04 marks)

5. a) List three (03) memory allocation strategies in OS. (03 marks)

b) Explain the difference between fixed partitioning and dynamic partitioning in memory management. (04 marks)

c) What are file system permissions? How are they used to control file access? (04 marks)

d) Analyze the efficiency and scalability of single-level and two-level directory structures in OS. (04 marks)

e) Explain the difference between precise and imprecise interrupts. (05 marks)

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